

NING Siyuan

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EDUCATION

Master of Science in Physics National University of Singapore Relevant Coursework: PC4236 Computational Condensed Matter Physics, PC5215 Numerical Recipes with Applications, PC5252 Bayesian Statistics and Machine Learning, PC5251 Applied Machine Learning, PC5253 Complex System Analysis and Modelling, PC5212 Physics of Nanostructures, PC4253 Thin Film Physics and Technology.	2023.08-2024.07 GPA什么原因不加上吗? 成绩差lol 3.8/5.0
Bachelor of Science in Physics, minor in Artificial Intelligence Shandong University Relevant Coursework: Material Simulation, Computational Physics, Group Theory, Solid State Physics, Film Physics and Techniques; (minor) Machine Learning, Neural Network and Deep Learning, Natural Language Processing.	2019.09-2023.06 985&211为何删除了? 已改

如果调整后不满两页, sections之间再多空一行, 比较舒服
感觉调整行距会好一些 已改

RESEARCH EXPERIENCE

MSc Physics Project supervised by Prof. QUEK Su Ying, National University of Singapore Worked on First-principles studies of quantum defect candidates in 2D WS ₂ • Developed a workflow to investigate 54 substitutional defects in monolayer WS ₂ by determining their ground-state structures, defect stability, spin configurations, and defect level positions. • Identified three spin-triplet defects, and examined their optimized local geometries as well as electronic properties. • Expedited setup and post-processing of calculations using bash scripts and VASP KIT toolkit. • Wrote a 60-page project thesis and present the results to a panel of examiners [PDF]	2023.08-2024.07 角标已改 不顺/不理解 不顺/不理解 不顺/不理解 拿了什么成绩呢? B+
Research Assistant under Prof. Zhang Peng, Shandong University Worked on Computational Analysis of Hydrogen Bond Vibrations of Ice III in the Far-Infrared Band • Calculated vibrational spectra of Ice III and its hydrogen-ordered counterpart ice IX, validated by experimental data. • Analyzed vibration normal modes of hydrogen bonds to explain the origin of far-IR vibrational bands. • Assessed structural deformations under high pressure in hydrogen bond lengths and included angles. • Authored first-authored paper published in Crystal [PDF]	2020.06-2022.11 截取自NUSMOD课程描述: Students will ..., submit a project thesis, and present the results to a panel of examiners. 有时大写有时小写? 已改 为何文件里有黄色高亮? 已改

我设想最理想的情况是每个project都做一个网站展示, 类似于学术海报, 这样的话可能CV中不用写太详细。

RELEVANT PROJECTS

Machine learning in Materials Informatics VASPKIT Workshop • Employed physics-informed featurization (Matminer) and Interpretable Machine learning techniques (SISSO, SHAP).	2024.07 之前提要写机构, 最后不写的原因是什么来着? 这个刚找到机构 Tencent Quantum Laboratory 其他不好写机构的都放到skill/self taught了
Sunspot Number Forecasting: Bayesian Neural Networks (BNNs) via MCMC [HTML] Final Project for PC5251 Applied Machine Learning, National University of Singapore • Built a 10x5x3 neural network, where parameters are initialized as distributions and updated via Langevin MCMC. • Obtained high-accuracy predictions with confidence intervals and conducted Gelman-Rubin convergence diagnostic. • Developed a highly-rated Jupyter notebook, recognized for its well-written and pedagogically accessible content.	2024.05 什么情况不能公开 不顺/不理解 已调整 不顺/不理解 已调整 评语内容不太会改
Deep Learning the Phase Transition: A CNN Approach to the 2D Ising Model [HTML] Research project in PC5212 Numerical Recipes, National University of Singapore • Used the Monte Carlo method to obtain representative sets of spin configurations for a bunch of temperatures. • Trained convolutional neural networks via PyTorch to classify configurations and predict the critical temperature.	2023.10

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Finite Element Analysis and Optimization of Permanent Magnet Motor

2023.03-2023.06

Capstone project in minor degree, Shandong University

● Established motor models using ANSYS Maxwell, defining solution domains, boundary conditions, and mesh subdivisions for basic motor performance simulation.

● Implemented multi-layer band refinement for calculate air-gap flux density and slot torque, followed by a multivariable parameter sweep of relevant structural parameters.

● Proposed structural optimization strategies aligned with output power metrics.

没懂哦

X
- Numerical Simulation on Quantum Size Effects in Nanomaterials

2022.03-2022.11

Research Project with Prof. Guan Chengbo, Shandong university
- 为什么这个没内容了?

RESEARCH SKILLS

这样表达很清晰! 但也要思考各个是不是那么值得被写出来(行业眼光衡量)。
- Computational Modeling Codes/Packages:

● DFT codes: VASP, CASTEP, SIESTA

● Automation & Processing: Vaspkit, Pymatgen

● Model Builder and Visualizer: Material Studio, VESTA

● Finite Element Analysis: ANSYS Workbench

Machine Learning Techniques:

● Frameworks: Scikit-learn, PyTorch, XGBoost, SiSSO

● Ensemble methods: Bagging, Boosting, Stacking

● Explainable tools: SHAP

Programming Language: Python (Numpy, Pandas), Shell (bash), MATLAB

Data Processing: Excel, Origin
- VOLUNTEER & TEACHING EXPERIENCE
- Volunteer Coordinator | Students' Union, School of Space Science and Physics, Shandong University

2020-2021

Led and coordinated volunteer activities at nursing homes and schools for children with autism in Weihai.

Teaching Assistant | Teaching Office of General Mathematics, Shandong University

2020-2021

Assisted in *Calculus* courses by preparing course materials, conducting tutoring, and grading.
- GRANTS & AWARDS
- Second-class Academic scholarship, Shandong University

2021-2023

Excellent Social Practice Research Report, Shandong University

2021

Volunteer Service Award, Shandong University

2020
- 上次讲过了

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THESIS PUBLICATION
- Ning, S.-Y.; Cao, J.-W.; Liu, X.-Y.; Wu, H.-J.; Yuan, X.-Q.; Dong, X.-T.; Li, Y.-N.; Jiang, Y.; Zhang, P. *Computational Analysis of Hydrogen Bond Vibrations of Ice III in the Far-Infrared Band*. Crystals 2022, 12, 910.

<https://doi.org/10.3390/cryst12070910>